

Task 3, part b answer:

Based on the output plots, NCC performed better than SSD.

Why?

SSD doesn't subtract the mean value of the patch to the pixels so when the second image is darker, SSD will only look at the absolute color differences in light color or dark color to determine if the 2 pixels are similar or not. This results in inaccuracy since the 2 pictures are the same but only have different brightness and SSD is not good with brightness.

NCC on the other subtracts the mean of the patch to every single pixel so it reduces the influence of brightness onto the pixels and therefore it has higher accuracy in matching pixels.